

Question:
What is the de Broglie wavelength of an electron with a kinetic energy of 1eV?

- Answer Choices:
- A. 2.48×10^{-8} cm
 - B. 5.92×10^7 cm/sec
 - C. 8.66×10^{-6} cm
 - D. 3.14×10^{-5} cm
 - E. 9.81×10^{-4} cm
 - F. 5.27×10^{-9} m
 - G. 1.23×10^{-7} cm
 - H. 4.14×10^{-10} m
 - I. 6.625×10^{-27} erg sec
 - J. 0.512×10^6 eV

Reasoning + Answer:

`<think>`
Use the classical electron energy-momentum relation for relativistic de Broglie wavelength:
$$\lambda = h / (\gamma (2mE)^{1/2} c),$$

where h is Planck's constant (in erg·s), m is the electron mass (9.11×10^{-28} g), c is the speed of light (3.00×10^{10} cm/s), and $E = 1$ eV = 1.60×10^{-12} erg.

Calculate the factor:
$$\gamma (2mE/c^2)^{1/2} = [(2 \times 9.11 \times 10^{-28} \times 1.60 \times 10^{-12}) / (3.00 \times 10^{10})^2]^{1/2} \approx 1.875 \times 10^{-23} \text{ g·cm/s}.$$

Then $\lambda = h / (\gamma (2mE)^{1/2} c) = (6.625 \times 10^{-27} \text{ erg·s}) / (1.875 \times 10^{-23} \text{ g·cm/s} \times 3.00 \times 10^{10} \text{ cm/s}) \approx 1.23 \times 10^{-7} \text{ cm}.$

Correct answer: G. 1.23×10^{-7} cm

`</think>`
`<answer>`
G. 1.23×10^{-7} cm
`</answer>`